

Probability and Bayesian Reasoning – random variables, conditional probability, independence, Bayes' rule, multiple evidence, normalization, Naive Bayes

CS 4880: Artificial Intelligence

Lecture 18 – October 7, 2026

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Today

- ▶ Course unit: Course sequence
- ▶ Big idea: Probability and Bayesian Reasoning – random variables, conditional probability, independence, Bayes' rule, multiple evidence, normalization, Naive Bayes
- ▶ Placeholder for announcements and context.

Learning Goals

- ▶ Define the key vocabulary for this lecture.
- ▶ Work through one representative example.
- ▶ Connect today's idea to the previous and next class meetings.

Placeholder

Replace these bullets with the final lecture material, examples, and in-class activities.

- ▶ Main concept 1.
- ▶ Main concept 2.
- ▶ Main concept 3.

Example or Activity

In class

Add a short worked example, code trace, proof sketch, discussion prompt, or board activity here.

Wrap-Up

- ▶ What should students be able to do after today?
- ▶ What should they practice before the next meeting?
- ▶ What question should they bring to class next time?